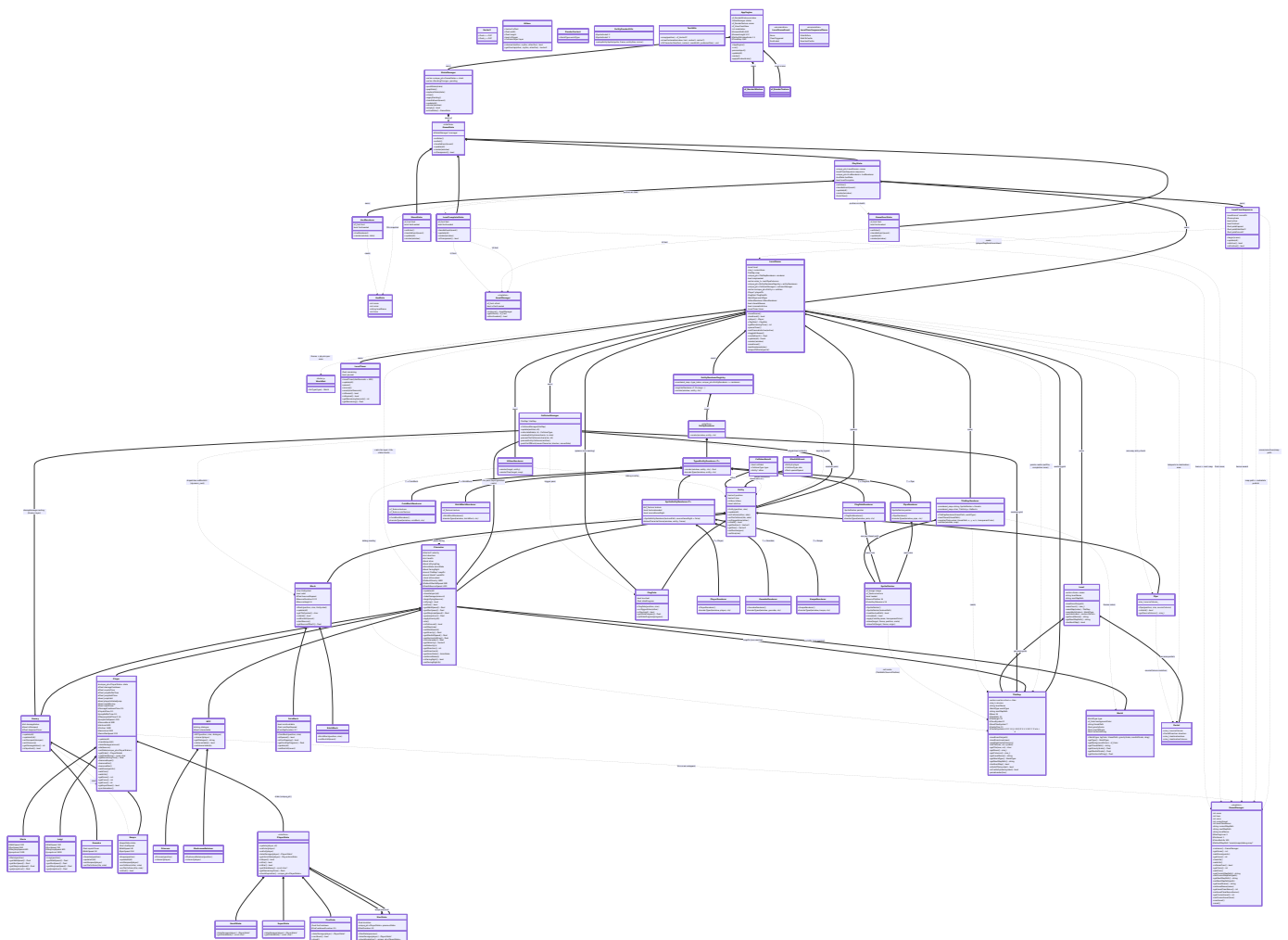


Full Project Class Diagram

Mermaid diagram covering all model, view, and controller classes (k.2.5). All three restructuring stages are reflected. Stage 1: model interface segregation. Stage 2: the goal castle is painted into the TileMap completion zone from a 21-symbol sheet (`TileMap::CastleSymbols` , registered by `TileMapRenderer` from atlas rows `y = 696/712` (tower) and `728/744/760` (base)); pipes spawn from 'p' / 'P' grid runs; tile collision treats castle symbols as ground/solid; and every static-image draw goes through the shared `SpritePainter` facade (the `Castle` entity and `CastleRenderer` are gone). Stage 3: the controller god-object split — the live level is `LevelScene` (update reports `Event s`), the flagpole clear play is `LevelClearSequence` , and `PlayState` is slimmed to state transitions, HUD snapshot and debug keys.

Companion doc: `CLASS_ENCAPSULATION_AND_CONNECTIONS.md` (per-class encapsulation + inter-class methods).



Enums

Enum	Values
CollisionType	None , Top , Bottom , Left , Right
CollisionLayer	Player , Enemy , Environment , Trigger
AnimState	Idle , Walk , Run , Jump , Fall , Die
PlayerAnimState	SmallIdle , SmallWalk , SmallRun , SmallJump , SmallFall , SmallDie , BigIdle , BigWalk , BigRun , BigJump , BigFall , FireIdle , FireWalk , FireRun , FireJump , FireFall , FireShoot , StarIdle , StarWalk , StarRun , StarJump , StarFall
KoopaState	Walking , ShellIdle , ShellSpinning
PortalDirection	Down , Up
WorldType	Overworld , Underwater , Castle
LevelScene::Event	None , ClearTriggered , RunEnded
LevelClearSequence::Phase	SlideToPole , WalkToCastle , ReachedCastle

Legend

Arrow	Meaning
< --	Inheritance
*--	Composition (owning)
o--	Aggregation (non-owning reference)
..>	Dependency / usage